TopoPatch invention disclosure

Completed responses to Invention Disclosure Format B

Reference form: VIT IPR&TTCELL, Document 02-IPR-R003, Issue 2 dated 01.02.2024. These continuation pages expand the ten fields in the supplied form and accompany its completed two-page copy.

1 Title of the invention

Validation and controlled commit of partial embedding model migrations

2 Field or area of invention

The invention concerns search systems that store document embeddings in a vector index. It focuses on replacing the embedding model for selected documents, checking the resulting search behaviour, and controlling when those changes become active. Possible uses include enterprise search and retrieval augmented generation systems.

Purpose of this disclosure

TopoPatch addresses a practical question: when only some documents have moved to a new embedding model, can the system check the effect on search results before making that change permanent? The prototype calculates the proposed mixed result, records the evidence and repeats the checks inside the commit operation.

The current evidence supports this control sequence as a local proof of concept. It does not establish that the sequence is inventive, or that selective migration gives the quality of a full upgrade at lower cost. The disclosure asks VIT to examine that narrower mechanism in light of the prior art and measured results.

The main embodiment removes migrated documents from the active old-model ranking. A separate experimental variant retains that ranking as a shadow signal. Their results and limitations are kept separate throughout this document.

3 Prior patents and publications

The references below show why model migration, selective indexing and validation cannot, by themselves, be treated as new. The final column identifies the narrower feature that needs comparison with each reference. These are candidate distinctions, not a finding that the invention is novel.

Reference	Relevant teaching	Proposed distinction to examine
[R1] US20240303516A1 Vector as a service HubSpot, 12 Sep 2024	Vector management and reindexing associated with embedding model changes.	Action-specific mixed-state retrieval evidence bound to source versions and checked again at commit.
[R2] US20250342179A1 Cloudera, 6 Nov 2025	Detecting dataset mutations and selectively updating affected vector-index content.	Migration selection based on measured retrieval effects of an A-to-B representation change, including unchanged source content.
[R3] Qdrant Migrate to a New Embedding Model Online documentation	Operational model migration using new vectors while maintaining service.	A recorded partial-index action and its retrieval effects, checked again against source state at commit.
[R8] US11610143B1 Amazon, 21 Mar 2023	Client-specific tests guide whether an updated machine-learning model is deployed. Claims 1 to 3 cover validation-based migration decisions.	Test the exact partial A/B document-membership change and bind it to source state at commit. This narrower distinction still needs examination.
[R9] US20260195572A1 9 Jul 2026	Localized embedding patches with a model that maps new embeddings back to the original space; see claim 1.	TopoPatch uses native-space rankings without a trained cross-space mapping. Generic local patching is not claimed as new.
[R10] SPFresh SOSP 2023	Incremental in-place vector-index updates using local partition rebalancing.	The prototype controls model migration through retrieval checks; it does not implement a new ANN repair method.
[R11] SQLite documentation	Standard isolation and atomic database transactions.	The retrieval-specific admission logic needs examination; the database transaction itself is established.

BEIR and SciFact [R4] are evaluation resources used in Section 8, not competing inventions. Live web documentation needs a dated archived version before it is relied on as pre-filing prior art.

The main question is whether testing a specific partial-index change and binding that result to a source-consistent commit adds a non-obvious technical contribution. The references make broad claims to validation, selective updates or rollback difficult to defend.

4 Summary and background of the invention

Changing an embedding model can change what a search system returns. During a partial upgrade, some documents use the old model and others use the new one. Combining their result lists can push an important document down the ranking even when that document has not been edited. Testing only the new vectors can miss this problem.

TopoPatch checks the proposed mixed index before activating the change. It uses the same ranking routine as the live search path and records which documents would move, their source hashes, the current migration epoch and the ranks of required results. The commit step checks that the evidence is still current and runs the retrieval checks again before updating active membership.

The feature proposed for examination is the link between a tested partial-index change, its measured effect on protected queries, and the checks that authorize the database update. The prototype rejects changes that fail those checks. It has not yet shown that a partial upgrade can match a full upgrade while using less total encoding work.

5 Objectives of the invention

• Find search regressions before a partial model upgrade becomes active.

• Keep enough information to identify the source and experiment state used in each preview.

• Reject a commit when the preview is stale, altered or fails the current checks.

• Count the target vectors computed during discovery and testing, including work on rejected actions.

• Allow a checked legacy route and restoration of the latest unchanged committed membership.

• Investigate whether this control process can reduce upgrade work without losing the benefit of the new model. This remains a research objective.

6 Working principle in brief

Let A be the old embedding model and B the new model. The controller computes B vectors for a proposed set of documents. Without changing live membership, it calculates the search results that would follow if those documents used B and the rest used A. It combines ranks from the two spaces and checks whether each protected result stays within its allowed rank. An optional development-query check also requires nDCG@10 to stay above a configured minimum.

A SHA-256 digest identifies the preview record. At commit, the controller starts a SQLite write transaction, checks the source snapshot and record digest, and repeats the retrieval checks. It then writes the target vectors, active membership, epoch and commit history together. If a check or write fails, the membership change is not committed. The digest detects changes to that record; it is not a digital signature or a tamper-proof audit system.

The prototype offers two ways to acquire candidates: a heuristic that examines a subset, and an exhaustive mode that computes every current B document vector. Strict admission refuses the heuristic evidence. Exhaustive mode establishes complete vector availability, but pays the full encoding cost and still does not guarantee that mixed results equal full-B results.

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7 Detailed description of the invention

Figure 1 shows the implemented sequence. A preview asks what the search results would be after a particular change. Computing vectors and recording the answer do not activate the change.

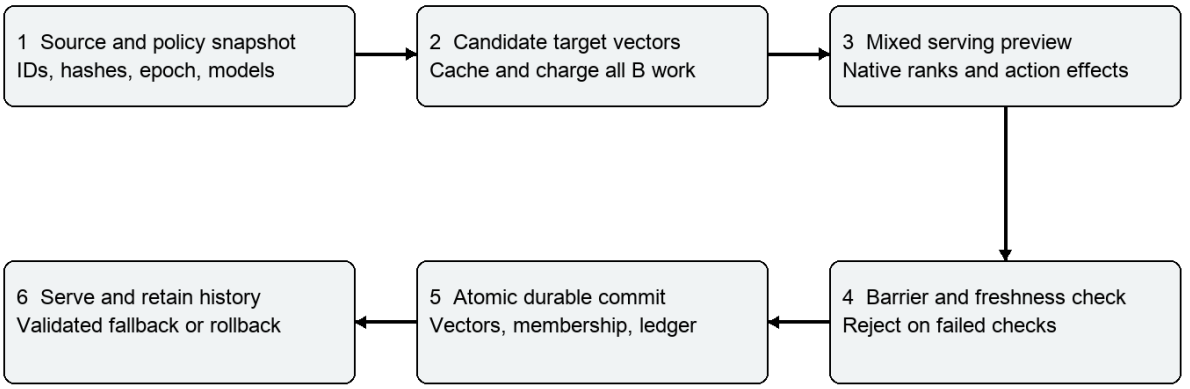


Figure 1 Implemented migration validation and commit sequence

Data and serving state

Each document stores its ID, text, version, source hash, A vector, optional B vector and active membership. The B cache uses the ID and source hash. Reopening checks model identifiers and fusion policy, assuming unchanged model files. Preview and event records retain the evidence.

In the default embodiment, migrated objects are excluded from the A result list. Reciprocal rank fusion uses 1/(60 + rank); ties favor B and then document ID. No cosine comparison is made between A and B vector spaces. The default fixture deliberately uses different vector dimensions, 128 and 256.

Witness selection and commit

Candidate discovery combines protected-query results, token overlap and required documents. The controller tests singletons and pairs over at most 24 candidates, then a protected-document bundle if needed. Each action record contains IDs, rank changes, pass status and object count. It chooses the smallest passing action among those tested, not a global or compute-cost optimum.

The snapshot covers source versions, membership, model identifiers, fusion and the development gate. Commit rechecks the snapshot, digest, source hashes and current retrieval constraints within a write transaction. Tests cover altered evidence, stale source or epoch, and write failure. Source changes invalidate B. Rollback restores the latest unchanged committed membership; the demonstration uses a server that serializes requests.

Example of an indirect ranking change

In the authored fixture, moving car-guide to B leaves q-car within its allowed rank but moves password-policy to rank 2 for q-policy, whose maximum is 1. The controller therefore rejects the action. Moving password-policy instead passes all four checks. This illustrates why the preview examines the mixed result list, including effects on documents outside the selected action.

Separate experimental embodiment

Continuity fusion retains all A vectors as a shadow ranking and weights A by 1 minus the migrated fraction and B by the migrated fraction. Its endpoints are legacy A and full B. This departs from the frozen A-removal embodiment and adds legacy storage and query work. ANN graph repair, distributed transactions and sublinear certified candidate discovery are not implemented.

8 Experimental validation results

The prototype passed 25 automated regression tests. The browser demonstration also exercised candidate analysis, rejection, commit and rollback. These checks support the behaviour described below. They do not establish novelty, production reliability or a reduction in migration cost.

Experiment	Measured result	Interpretation
Authored 40-document fixture 4 protected queries	14 B document encodings; 1 migrated object; 13 rejected singleton actions; 0 protected violations.	The gate makes an observable decision. Held-out full-B top-5 overlap is 0.80, below legacy 0.95.
Full fixture trajectory	40 safe commits; 40 B document encodings; full membership migration.	The controller can finish this fixture. There is no final encoding saving.
Exhaustive strict evidence 40-document fixture	All 40 B vectors materialized and charged; an admissible commit completed.	Complete candidate coverage is available at full cost. Candidate enumeration remains capped.
Source and transaction tests	Stale source, stale epoch, altered preview and injected write failure rejected.	Active membership is protected under the tested local SQLite failure cases.

Public dataset experiments

SciFact samples contain 160 documents, selected by retaining relevant documents for sampled queries and adding distractors. This relevance-aware selection can make evaluation easier and is not the full BEIR benchmark. The table reports held-out nDCG@10; higher values indicate better relevance ranking. L6 and L12 refer to the cached all-MiniLM sentence-transformer models.

Variant and split	A / selective / B	B charged / migrated	Result
L6 to L12, RRF 6 protected, 12 held-out	.887 / .526 / .893	80 / 6	300 candidate previews rejected; 0 protected violations; 15% full-B top-10 envelope miss rate.
L6 to L12, continuity 6 protected, 12 held-out	.887 / .887 / .893	80 / 1	Preserved A quality without measured improvement. Exploratory split reused during redesign.
Lexical to L12, continuity 6 protected, 6 development, 6 held-out	.384 / .384 / .898	83 / 1	Development floor passed; target-quality progress was not achieved. Envelope miss rate 28.3%.

Full-B evaluation additionally encoded all 160 documents in a separate ledger for each run. It was not supplied to action selection. Historical runs differ in constraints and splits; they are not a controlled ablation. The random fixture baseline matches migrated-object count, not discovery compute. No equal-budget superiority or statistically significant quality gain is claimed.

The experiments support a limited conclusion: the controller can reject actions that fail its declared checks. Passing those checks did not reliably preserve quality on other queries. The intended quality and cost advantage remains unproven.

8 Experimental validation screenshots

Actual local browser captures from the authored lexical fixture. Screenshots illustrate the workflow; numeric results and failure checks are supported by the JSON exports and test log.

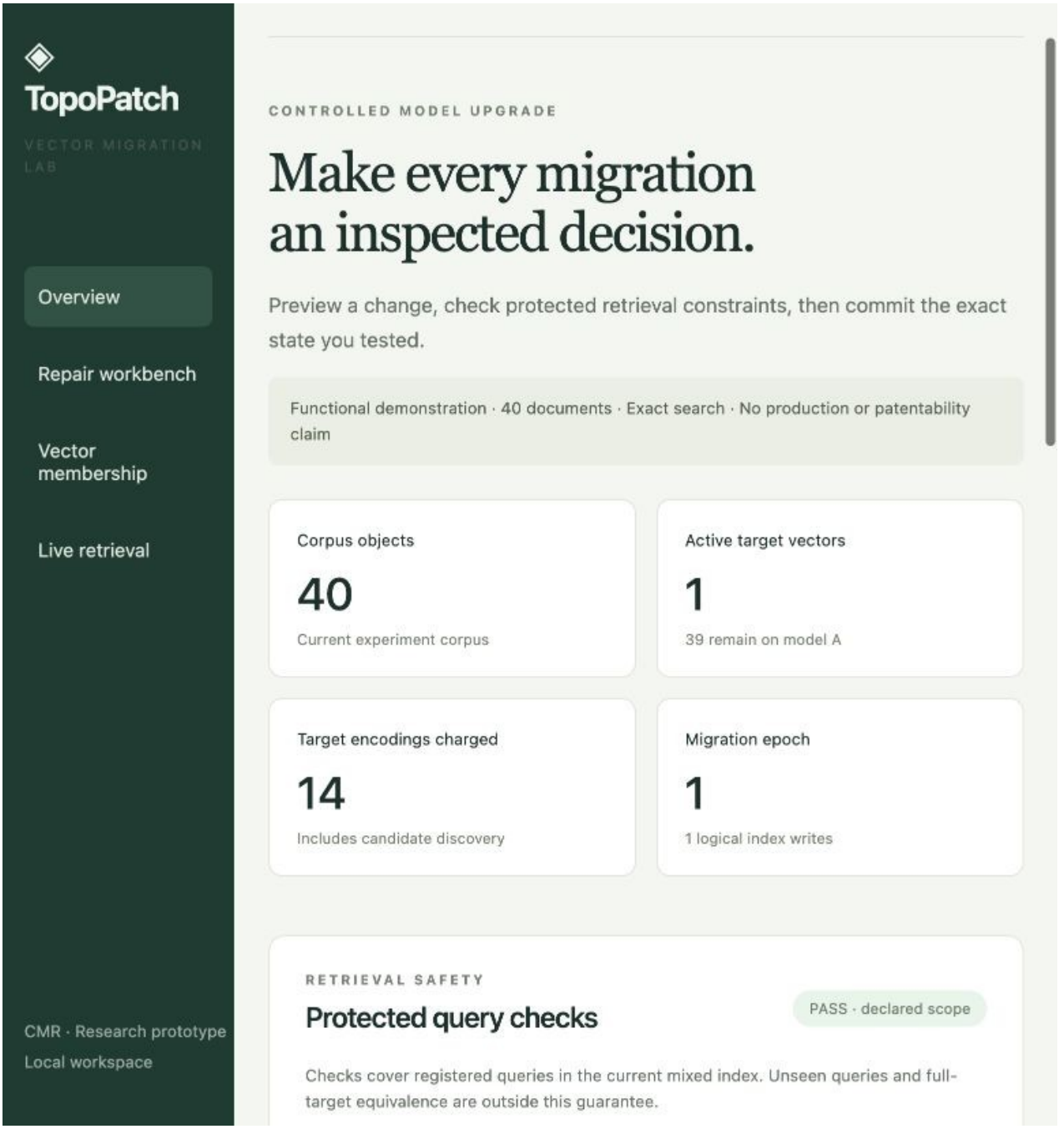


Figure 2 Successful commit: 1 active B object, 14 target encodings and epoch 1.

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8 Experimental validation screenshots continued

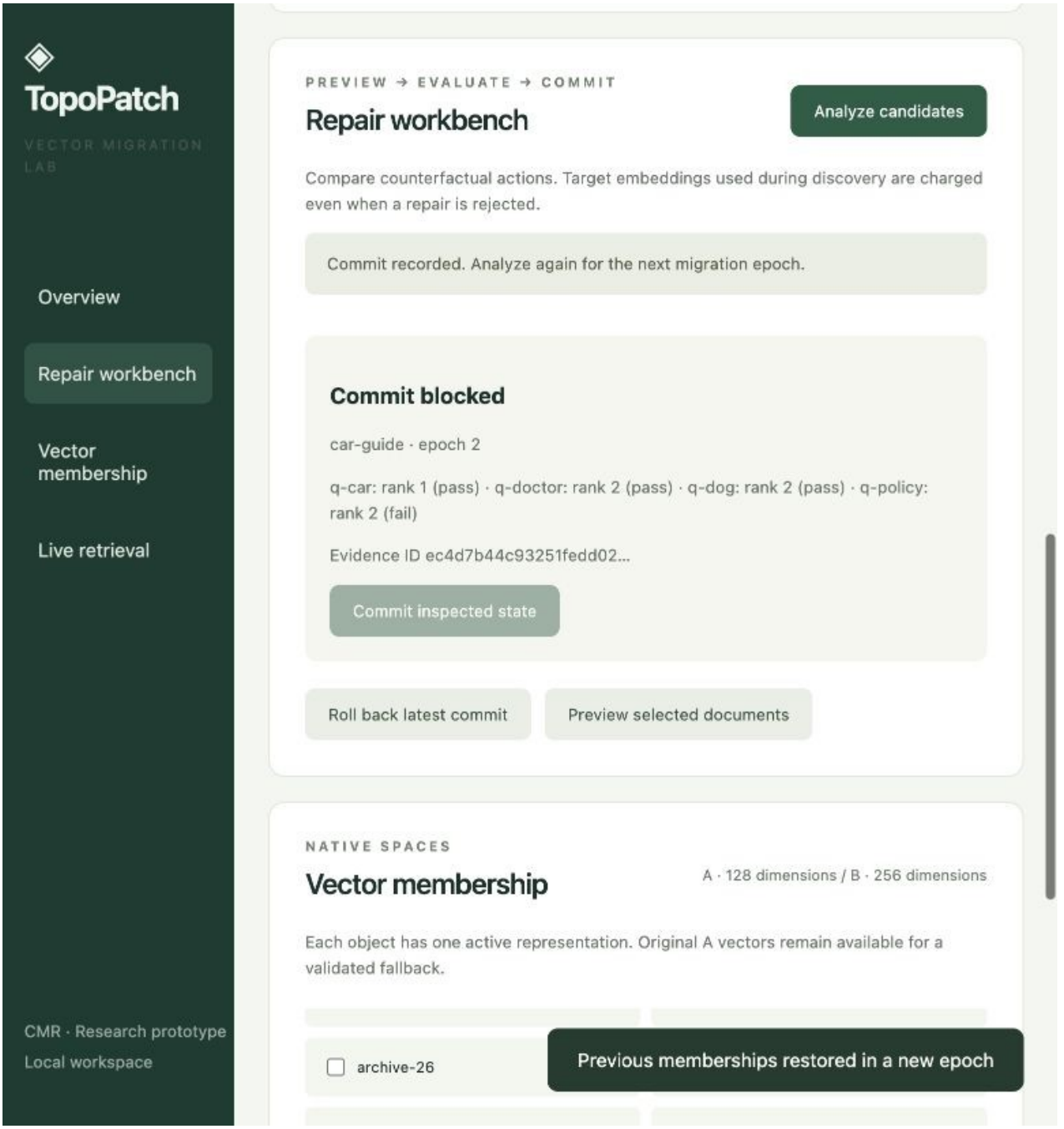


Figure 3 After rollback to epoch 2, the car-guide preview fails q-policy and commit is disabled.

The full-resolution screenshots are included in the evidence package. No screenshot represents a neural benchmark or a production deployment.

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9 Aspects of the invention needing protection

We ask the IPR Cell to examine the following sequence as a possible combined contribution. The proposed scope is limited to the concrete migration process described here:

• Calculate the mixed search state for a selected set of B vectors, including removal of those documents from the active A ranking, before committing the change.

• Record the effect of that action on protected result ranks, together with document IDs, source digests and experiment state, and use the record to choose an admissible action.

• Identify the preview by a digest of its tested vectors and recorded state. Within the write transaction, reject stale or altered evidence and recheck the current retrieval constraints before changing membership.

• In the strict mode, require current target-vector coverage before admission. The implemented exhaustive fallback computes all vectors and records that full cost.

• Keep an inspectable event history, restore the latest unchanged committed membership when requested, and allow a legacy route only for a query that has passed its legacy check.

The intended effect is to stop an index change that fails the inspected retrieval constraints or relies on stale source evidence. The local tests support that behaviour. Whether the combination adds an inventive step over model validation and standard transaction control remains unresolved. Rank fusion, hashing, caching, SQLite atomicity and the dashboard are supporting techniques. The exhaustive fallback is not claimed to save encoding work.

A patent professional should assess the combined mechanism under the applicable computer-related invention requirements, including Section 3(k) of the Indian Patents Act [R7]. These proposed protection points are disclosure material, not final patent claims or a conclusion of patentability.

10 Technology readiness level

Selected level: [X] TRL 3 — Experimental proof of concept

[ ] TRL 1 [ ] TRL 2 [X] TRL 3 [ ] TRL 4 [ ] TRL 5
[ ] TRL 6 [ ] TRL 7 [ ] TRL 8 [ ] TRL 9

The project has a working local prototype and experimental evidence for its validation and commit sequence. We propose TRL 3 because the central mechanism has been demonstrated as a proof of concept. The faculty should confirm this assessment using the VIT guide [R6].

The full frozen design has not reached the validation claimed for TRL 4. An efficient certified candidate method, ANN topology repair and independent evidence of the intended quality and cost benefit are still missing.

Faculty review focus

We request faculty review of the mechanism and its differences from the listed references. VIT requires disclosure through Parts A and B and may use domain experts before referring an approved case to a patent attorney [R5]. This document completes Part B; it is not a patent application or an approval to file.

Evidence and reproduction

Run the offline demonstration from the project directory with Python 3.11 or later:

python3 -m topopatch serve --port 8767 --db artifacts_private/faculty-demo.sqlite

Open http://127.0.0.1:8767. Analyze candidates, inspect the rejected car-guide action, inspect and commit password-policy, export evidence, then roll back. Use a previously unused database path for an initial-state demonstration.

Strict exhaustive demonstration: add --evidence exhaustive and use another fresh database. It should charge all 40 target vectors. Strict unresolved demonstration: use --evidence strict; analysis and commit must refuse unbounded evidence.

Regression suite: python3 -m unittest discover -s tests -v
Fixture benchmark: python3 -m topopatch benchmark

Evidence file in artifacts_private	Purpose
test-results.txt	25-test regression result
benchmark.json and trajectory.json	Fixture single-step and full trajectory ledgers
exhaustive-evidence.json	Complete candidate coverage and strict commit
scifact-rrf-strict-ranks.json	Frozen-style RRF experiment reported in Section 8
scifact-continuity-strict-ranks.json	Shadow-A continuity comparison
scifact-lexical-to-neural-progress-gated.json	Separated development and held-out query run
final-demo-evidence.json	Export from the screenshot demonstration
final-overview.jpg and final-blocked-detail.jpg	Unmodified browser captures

The archive includes a SHA-256 manifest for traceability. A hash identifies the packaged bytes; it is not a notarized timestamp or proof of authorship. Historical JSON files preserve their original measurements and model identities.

References

[R1] HubSpot. Vector as a service. US20240303516A1.
https://patents.google.com/patent/US20240303516A1/en

[R2] Cloudera. Methods and apparatus to manage input data sets to reflect dataset mutations for GenAI and RAG applications. US20250342179A1.
https://patents.google.com/patent/US20250342179A1/en

[R3] Qdrant. Migrate to a New Embedding Model.
https://qdrant.tech/documentation/tutorials-operations/embedding-model-migration/

[R4] BEIR benchmark repository and SciFact archive.
https://github.com/beir-cellar/beir

[R5] VIT. Intellectual Property Rights and Technology Transfer Policy and Guidelines. Amendment 16 April 2024.
https://vit.ac.in/files/IPR-and-Technology-Transfer-Policy.pdf

[R6] VIT. A guide on Technology Readiness Levels.
https://vit.ac.in/wp-content/uploads/2025/08/A_guide_on_Technology_Readiness_Levels.pdf

[R7] IP India. Patents Act 1970, Section 3.
https://ipindia.gov.in/acts/patent-act-1970/section-3

[R8] Amazon. Client-specific validation for migrating to a version of a machine learning model using client-supplied test data. US11610143B1.
https://patents.google.com/patent/US11610143B1/en

[R9] Embedding Patches for Localized Improvement of Embeddings. US20260195572A1.
https://patents.justia.com/patent/20260195572

[R10] Xu et al. SPFresh. SOSP 2023. Author laboratory publication page.
https://ustc-mlsys.github.io/publications/spfresh-sosp-2023/

[R11] SQLite. Isolation in SQLite.
https://sqlite.org/isolation.html

Online references checked 15 October 2026. Source dataset archive: data/scifact.zip. The original project specification is retained in doc/.